



The Solar Cook Book

Professional Solar Guide for Jamaica

From phone-charging kits to
10kW+ home arrays

Learn to size, wire, and build
solar systems

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Welcome to Solar DIY

Solar energy is no longer just for large corporations or wealthy homeowners. With falling panel prices and improved battery technology, anyone can build their own solar power system.

Getting Started

This guide covers systems from 100W portable kits to 10kW+ home arrays. All examples are tailored for Jamaica's climate with 5.0-5.5 peak sun hours daily.



What You'll Learn

Panels

Monocrystalline
vs poly,
wattage sizing,
wiring configs

Batteries

LiFePO4 vs
lead-acid,
capacity math,
runtime

Controllers

PWM vs MPPT,
voltage
matching,
efficiency

Inverters

Pure sine wave
sizing, surge
loads, AC
distribution

Small Systems

(100W-500W)

Ideal for phone charging, laptop charging, LED TVs, and small electronics.

System	Use Case	Cost Estimate
100W Portable Kit	Charge 2 phones + 1 laptop daily; run LED TV 2 hours	50-50 USD
200W Starter	Adds headroom for cloudy days. Great for sheds and RVs	00-50 USD
400W Cabin Kit	Small off-grid setup with inverter for	00-00 USD

	lights, fans, charging	
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Pro Tip: Start small. A 100W kit teaches you the basics without a big investment. You can always expand later.

Medium Systems

(500W-2kW)

Perfect for Starlink Mini, larger TVs, multiple laptops, and home office setups.

System	Use Case
800W Office	Power 2 laptops + TV + router during outages. Great for remote work.
1.2kW Hybrid	Adds battery backup for evening use. Supports JPS net-billing if grid-tied.
1.8kW Expandable	Room to add more panels later. Can handle small appliances plus devices.

Key Components

Panels:

400W-1kW
monocrystalline
array

Battery: 12V

100Ah-200Ah
LiFePO4 bank

Controller:

40A MPPT
charge
controller

Inverter:

1000W-
2000W pure
sine wave

Large Systems (2kW-10kW+)

Full home power: multiple TVs, laptops, Starlink Mini, fridge, and all devices.

System	Use Case
3kW Home Backup	Covers essential circuits: lights, wifi, fridge, TV, and laptop loads.
5kW Whole-Home	Run most household circuits. Ideal for frequent outages or full off-grid.
10kW+ Estate	Multiple inverters, large battery bank, commercial-grade installation.

Professional Installation by Nash Technologies: Large systems require certified electricians. Nash Technologies

provides full installation, JPS net-billing support, 15-year warranties, free site assessments, and custom system design for homes and businesses across Jamaica.



Safety First

Electrical Safety Warning: Solar systems produce high voltage DC that can cause serious injury or fire. Follow NEC code, use properly sized wire, install fuses/breakers on all conductors, and hire a licensed electrician for grid-tied connections.

Wire Sizing & Fusing

Undersized wiring overheats. Size conductors for **1.25× expected current**. Fuse every positive conductor at the battery and between components. Use marine-grade or solar-rated cable for outdoor runs.

NEC Requirements

Disconnects:

DC and AC
disconnects
required

Grounding:

All frames and
enclosures
grounded

Labeling:

Warning signs
at all junction
boxes

Inspections:

JPS inspection
for grid-tied
systems

DIY Device Charging Systems

Step-by-step build guides for charging phones, laptops, Starlink Mini, and TVs with solar.

Phone & Tablet Charging Kit

- **Panel:** 20W monocrystalline
- **Battery:** 12V 7Ah LiFePO4
- **Controller:** 10A PWM
- **Output:** 5V USB-A ports
- **Runtime:** 2-3 phones per day

Laptop Charging Station

- **Panel:** 100W monocrystalline

- **Battery:** 12V 50Ah LiFePO4 (~640Wh)
- **Controller:** 20A MPPT
- **Inverter:** 300W pure sine wave
- **Runtime:** 2-4 hours laptop use

More Device Setups

Starlink Mini Power

- **Panel:** 120W monocrystalline
- **Battery:** 12V 100Ah LiFePO4
- **Controller:** 20A-30A MPPT
- **Inverter:** 300W pure sine wave
- **Runtime:** 6-10 hours off-grid

TV & Entertainment

- **Panel:** 150W monocrystalline
- **Battery:** 12V 100Ah LiFePO4
- **Controller:** 20A MPPT
- **Inverter:** 500W pure sine wave
- **Runtime:** 3-6 hours for 55" LED

Solar & Electrical

Glossary

Simple definitions for common terms on solar labels, invoices, and JPS net-billing paperwork.

Term	Definition
Watt (W)	Unit of power. A 100W bulb uses 100 watts while on.
Kilowatt (kW)	1,000 watts. A 5 kW system produces 5,000 watts at peak sun.
Kilowatt-hour (kWh)	Unit of energy. JPS bills in kWh.
Volt (V)	Electrical pressure. Common solar banks: 12V, 24V, 48V.
Ampere (A)	Current flow. Wiring and fuses are sized by amps.

Amp-hour (Ah)	Battery capacity. A 100Ah battery delivers 5A for 20 hours.
PSI	Pounds per square inch. Used for water pressure and pipe fittings.
BTU	British Thermal Unit. Heat energy; useful for water-heating sizing.
Peak Sun Hours	Hours of 1,000 W/m ² irradiance. Jamaica averages 5.0–5.5.
Inverter	Converts DC battery power to AC for appliances.
Charge Controller	Regulates panel-to-battery charging to prevent overcharge.
Grid-Tied	Connected to JPS lines. Allows net billing for excess energy.

Solar Calculator

Estimate daily energy production and system cost for Jamaica conditions.

Formula: $\text{Panel Watts} \times \text{Count} \times \text{Sun Hours} \times \text{Efficiency} \div 100$

Input	Value
Panel Watts	400W
Panel Count	4
Sun Hours (Jamaica)	5.5
Efficiency	75%
Daily Output	8,800 Wh
Estimated Cost	,280 USD

Use this formula to size any system. Adjust panel count and wattage to match your energy needs.

Solar Parts Finder

Select a system to see the full bill of materials with local-market pricing in JMD.

Component	Phone/ Tablet	Laptop Station	Starlink Mini
Panel	20W — ,000 JMD	100W — 2,000 JMD	120W — 4,000 JMD
Battery	12V 7Ah LiFePO4 — ,500 JMD	12V 50Ah LiFePO4 — 8,000 JMD	12V 100Ah LiFePO4 — 5,000 JMD
Controller	10A PWM — ,500 JMD	20A MPPT — ,000 JMD	20A–30A MPPT — 4,000 JMD
Inverter	—	300W — 2,000 JMD	300W — 2,000 JMD

Other	5V USB-A — ,200 JMD Wiring — ,500 JMD	Wiring + breakers — ,000 JMD	DC-DC converter — ,000 JMD
Total	~7,700 JMD	~3,000 JMD	~0,000+ JMD

For full home systems and custom builds,
contact Microtuff Solutions for a detailed
quote.

Nash Technologies

Professional Solar Installation
Partner

Trusted Installer: Nash Technologies is a certified solar installation company serving all 14 parishes of Jamaica. Specializing in residential, commercial, and industrial solar power systems.

Services

Residential Solar

Complete home solar systems from 3kW to 10kW+

Commercial Installation

Office buildings, warehouses, and retail spaces

JPS Net-Billing

Full paperwork and interconnection support

System Design

Custom layouts, equipment selection, and load analysis

Maintenance

Panel cleaning, inverter service, battery checks

Warranty Support

15-year warranties on workmanship and equipment

Why Choose Nash Technologies?

- **Certified Installers:** Trained and licensed by Jamaica Public Service (JPS)
- **Free Site Assessments:** No-obligation evaluation of your property and energy needs
- **Quality Equipment:** Only tier-1 panels, LiFePO4 batteries, and MPPT controllers
- **Local Support:** Based in Mandeville, serving all parishes
- **15-Year Warranties:** Peace of mind on every installation
- **JPS Interconnection:** Handle all permits, inspections, and net-billing agreements

Contact Nash Technologies

Phone:

1(876)
359-9983

Service**Area:**

All 14
Parishes,
Jamaica

Headquarters:

Mandeville,
Jamaica

Email:

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Ready to go solar?

Call or WhatsApp Nash
Technologies today for a free
consultation and site assessment.

Need Help?

Professional installation, system design,
and JPS net-billing support across
Jamaica.

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Certified solar
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**Ready to start your solar
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Contact us for a free consultation
and site assessment. We serve all
14 parishes across Jamaica.

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This guide is for educational purposes.
Always consult a licensed electrician for
installation.